

Lab 11: Faraday's Law

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Pre-lab Questions

1. The first graph shows that the magnetic flux has maximum magnitude when the center of the magnet is at the center of the loop. What is the induced emf at this instant?

The induced emf when the flux reaches its maximum is 0 V. Since the magnetic flux reaches an extremum at this instant, its derivative with respect to time is zero. By Faraday's law, the induced emf is equal to the negative rate of change of magnetic flux, so the emf is zero at this instant. Alternatively, we can reason that at this instant, the magnetic flux through the loop goes from increasing to decreasing in magnitude, so there is a point at which the flux is not changing at all, which is where the emf is zero.

2. In the simulation, positive flux corresponds to field lines passing from left to right through the loop. Would either of the graphs change if the magnet was turned around and the south pole passed through the loop first?

The graphs would be flipped. Magnetic field lines on a bar magnet emerge from the north pole and enter the south pole. When the north pole passes through the loop from right to left, the magnetic field emerging from the north pole points from right to left, causing a negative magnetic flux. If we flip the magnet, the south pole would pass through the loop first, and the magnetic field lines would point from left to right, causing a positive magnetic flux. The emf graph would also flip because the emf is proportional to the rate of change of flux.

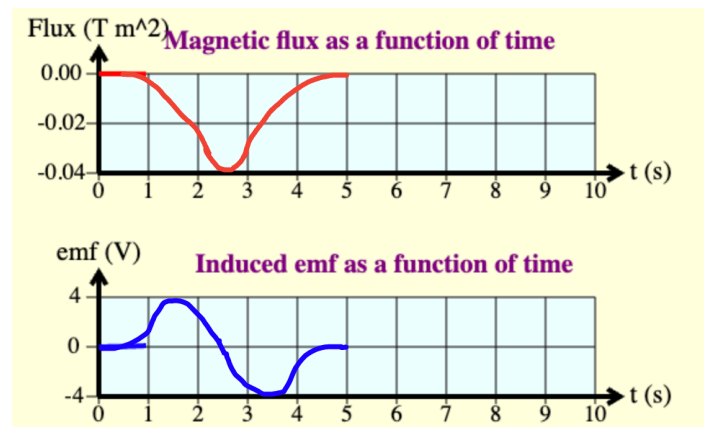
3. If the velocity of the magnet is doubled, would the graph of magnetic flux change?

The magnetic flux graph would be "compressed" horizontally by a factor of two. That is, the time at which we reach maximum magnetic flux magnitude will be 2.5 s instead of 5 s, and the width of the peak will be half as wide. However, the height of the peak will not change.

4. Would the graph of induced emf change?

The emf graph would also be compressed horizontally by a factor of two, but it would also be stretched vertically by a factor of two. This is because the emf is proportional to the rate of change of flux, so doubling the velocity doubles the slope of the flux graph at every point, doubling the emf.

Sketch of graphs for answers to questions 3 and 4:



Part 1: Time-dependent Magnetic Flux from Permanent Magnet Dropped through a Coil of Wire

Questions

1. When the magnet is dropped through the coil with the south pole down, why do we see two peaks and why are the two peaks in opposite directions?
2. How do the two peaks change if the magnet is dropped with the north pole down?
3. Why is the second peak higher than the incoming peak?
4. What does the area under the curve Potential vs. Time represent?
5. Are the two areas the same? Why or why not?
6. How does the magnitude of the induced EMF depend on the speed of the magnet?
7. How does the magnitude of the induced EMF depend on the strength of the magnet?
8. How would the magnitude of the induced EMF change if two magnets were taped together so that:
 - a. both south ends were together?
 - b. south end of one magnet is taped to north end of the other?
9. Would the experiment give similar results if the magnet would be stationary and the coil would be dropped?
10. Using your data and the known geometries, estimate the strength of the magnetic field of the two bar magnets? Assume magnetic field is confined to the magnet. What extra measurements do you need? Compare your estimate with an actual measurement of the magnetic field using the magnetic field sensor.

Part 2: Primary and Secondary Coils

Magnetic Flux Derivation

Let f be the frequency of the function generator, so $\omega = 2\pi f$ is the angular frequency of the potential. Let V_0 be the amplitude of the potential created by the function generator.

The function generator creates a potential $V(t)$ which follows:

$$V(t) = V_0 \sin(\omega t) = V_0 \sin(2\pi f t).$$

Let the primary solenoid have turn density n_1 , resistance R , and cross-sectional area A .

The current through the primary coil is given by Ohm's law:

$$I(t) = \frac{V(t)}{R} = \frac{V_0 \sin(2\pi f t)}{R}.$$

The magnetic field inside the solenoid is:

$$B(t) = \mu_0 n_1 I(t) = \frac{\mu_0 n_1 V_0 \sin(2\pi f t)}{R}.$$

The magnetic flux through one turn of the coil is:

$$\Phi_B(t) = B(t)A = \frac{\mu_0 n_1 A V_0 \sin(2\pi f t)}{R}.$$

Since we are given $n_1 = N_1/L$, where N_1 is the number of turns and L is the length of the solenoid, as well as $A = \pi r^2$ for the cross-sectional area (our solenoid is circular with radius r), we can rewrite the magnetic flux as:

$$\Phi_B(t) = \frac{\mu_0 \pi r^2 N_1 V_0 \sin(2\pi f t)}{LR}.$$

Questions

1. Write the expression for the magnetic flux in the primary/inner coil.
2. Use the expression from (1) to derive an expression for the induced emf in the secondary/outer coil.

The secondary coil will experience the same magnetic flux over its N_2 turns, so by Faraday's law, the emf induced in the secondary coil is:

$$\begin{aligned} \mathcal{E}(t) &= -N_2 \frac{d\Phi_B}{dt} = -N_2 \frac{d}{dt} \frac{\mu_0 n_1 A V_0 \sin(\omega t)}{R} = -\frac{\mu_0 n_1 N_2 A V_0 \omega \cos(\omega t)}{R} \\ &= -\frac{\mu_0 \pi r^2 N_1 N_2 V_0 2\pi f \cos(2\pi f t)}{LR} \\ &= -\frac{2\mu_0 \pi^2 r^2 N_1 N_2 V_0 f \cos(2\pi f t)}{LR}. \end{aligned}$$

3. Use the expression from (2) to interpret the two graphs you produced in Part II.
4. Does the data from your measurements match your theoretical expressions?